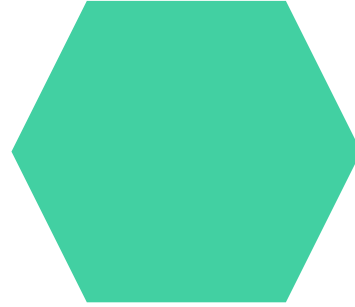
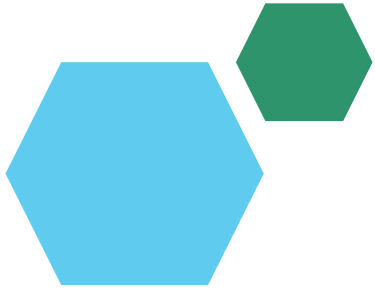




Project Title -Seasonal Agriculture Performance Analysis

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AICTE STU ID :

STU6a0c7ba72c2c01779202983



PROBLEM STATEMENT

- Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions. As a result, agricultural performance may differ from one season to another. However, raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed in different seasonal conditions. The problem is to analyze the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data.



Project Description

Seasonal Agriculture Performance Analysis is a data analysis project that examines agricultural activities across different seasons, geographical regions, and farming conditions.

The project analyzes factors such as **rainfall, temperature, humidity, soil conditions, crop yield, resource usage, water efficiency, production costs, revenue, and profit** to understand how agricultural performance varies across **Kharif, Rabi, and Zaid seasons**.

The analysis uses **data cleaning, statistical analysis, comparisons, correlations, and visualizations** to identify seasonal patterns, relationships, variations, and unusual observations.

The final outcome is to generate **evidence-based insights and recommendations** that can support better seasonal agricultural planning and resource management.

WHO ARE THE END USERS?

- **END USERS**
- **Farmers** - Understand seasonal crop performance, resource usage, and profitability.
- **Agricultural Planners** - Use seasonal trends and regional comparisons for better planning.
- **Agricultural Departments** - Analyze seasonal agricultural performance across regions.
- **Agricultural Researchers** - Study relationships between environmental conditions and crop outcomes.
- **Agribusiness & Market Analysts** - Understand seasonal changes in production, costs, revenue, and profit.
- **Agricultural Consultants** - Provide data-driven recommendations for improving farm management.

Technology Used

- **Python** - Main programming language for data analysis
- **Pandas** - Data loading, cleaning, manipulation, and analysis
- **NumPy** - Numerical operations and data processing
- **Matplotlib** - Creating charts and visualizations
- **Seaborn** - Statistical and comparative visualizations
- **SciPy** - Statistical analysis and ANOVA testing
- **Google Colab** - Development and execution environment
- **Jupyter Notebook** - Documentation and presentation of the analysis



RESULTS

1. How does agricultural performance vary across seasons?

We can compare Yield, Production, Revenue, Cost, profit, Water efficiency, Disease risk

```
season_performance = df.groupby('Season').agg([
    'Yield_Tonnes_Ha': 'mean',
    'Production_Tonnes': 'mean',
    'Revenue_INR': 'mean',
    'Total_Cost_INR': 'mean',
    'Profit_INR': 'mean',
    'Water_Efficiency_t_per_1000m3': 'mean',
    'Disease_Pest_Risk_pct': 'mean'
]).round(2)
```

```
print(season_performance)
```

Season	Yield_Tonnes_Ha	Production_Tonnes	Revenue_INR	Total_Cost_INR
Kharif	5.64	46.31	710719.06	531804.41
Rabi	5.08	41.49	601526.05	513836.58
Zaid	4.67	38.89	519171.90	543976.73

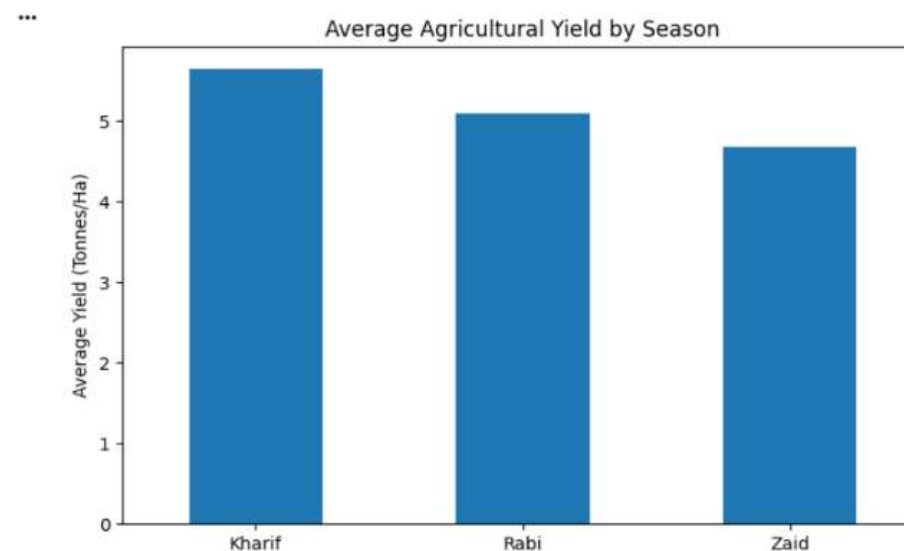
Season	Profit_INR	Water_Efficiency_t_per_1000m3	Disease_Pest_Risk_pct
Kharif	178914.65	5.89	54.47
Rabi	87689.47	5.19	40.48
Zaid	-24804.82	4.41	38.22

Bar chart comparing Yield / Revenue / Profit across Kharif, Rabi and Zaid.

Kharif showed the highest average agricultural performance, followed by Rabi and Zaid. Kharif recorded higher average yield, revenue, profit and water efficiency compared with the other seasons.

```
season_performance['Yield_Tonnes_Ha'].plot(
    kind='bar',
    figsize=(8,5),
    title='Average Agricultural Yield by Season'
)

plt.xlabel('Season')
plt.ylabel('Average Yield (Tonnes/Ha)')
plt.xticks(rotation=0)
plt.show()
```



RESULTS

2. What major seasonal patterns can be observed?

Here we don't look at only one variable. We look at several variables together.

```
season_pattern = df.groupby('Season').agg({
    'Rainfall_mm': 'mean',
    'Avg_Temperature_C': 'mean',
    'Humidity_pct': 'mean',
    'Sunlight_Hours_Day': 'mean',
    'Soil_Moisture_pct': 'mean',
    'Yield_Tonnes_Ha': 'mean',
    'Profit_INR': 'mean',
    'Water_Used_m3': 'mean'
}).round(2)

print(season_pattern)
```

Season	Rainfall_mm	Avg_Temperature_C	Humidity_pct	Sunlight_Hours_Day
Kharif	852.08	28.45	71.81	6.79
Rabi	436.00	23.49	57.89	7.59
Zaid	299.42	31.04	52.01	8.18

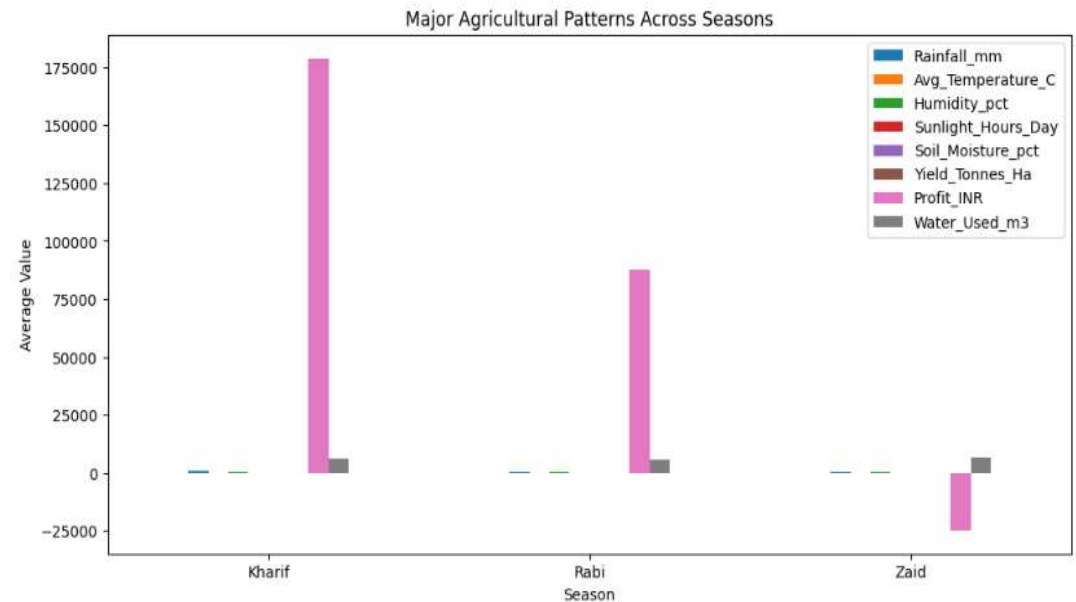
Season	Soil_Moisture_pct	Yield_Tonnes_Ha	Profit_INR	Water_Used_m3
Kharif	31.20	5.64	178914.65	6102.20
Rabi	24.05	5.08	87689.47	5846.99
Zaid	19.17	4.67	-24804.82	6419.89

The visualization highlights clear seasonal differences in **rainfall, temperature, soil moisture, sunlight, and crop yield**, showing that agricultural conditions change considerably between seasons.

Major patterns can be identified by comparing agricultural performance, environmental conditions, resource usage, and economic outcomes across different seasons.

```
season_pattern.plot(
    kind='bar',
    figsize=(12,6),
    title='Major Agricultural Patterns Across Seasons'
)

plt.xlabel('Season')
plt.xticks(rotation=0)
plt.ylabel('Average Value')
plt.show()
```



RESULTS

3. Which characteristics change between seasons?

```
characteristics = [  
    'Rainfall_mm',  
    'Avg_Temperature_C',  
    'Humidity_pct',  
    'Sunlight_Hours_Day',  
    'Soil_pH',  
    'Soil_Moisture_pct',  
    'Nitrogen_kg_ha',  
    'Phosphorus_kg_ha',  
    'Potassium_kg_ha',  
    'Fertilizer_kg_ha',  
    'Pesticide_Litre_ha',  
    'Seed_Quality_Score'  
]  
  
characteristic_comparison = df.groupby('Season')[characteristics].mean().round(2)  
  
print(characteristic_comparison.T)
```

Season	Kharif	Rabi	Zaid
Rainfall_mm	852.08	436.00	299.42
Avg_Temperature_C	28.45	23.49	31.04
Humidity_pct	71.81	57.89	52.01
Sunlight_Hours_Day	6.79	7.59	8.18
Soil_pH	6.73	6.67	6.69
Soil_Moisture_pct	31.20	24.05	19.17
Nitrogen_kg_ha	120.02	120.40	120.53
Phosphorus_kg_ha	58.10	57.31	58.13
Potassium_kg_ha	105.30	103.93	105.42
Fertilizer_kg_ha	187.08	185.41	184.64
Pesticide_Litre_ha	5.08	5.02	5.17
Seed_Quality_Score	0.82	0.83	0.82

Several environmental characteristics change significantly. Average temperature increases from 23.49°C in Rabi to 31.04°C in Zaid, while soil moisture decreases from 31.20% in Kharif to 19.17% in Zaid.

RESULTS

4. What differences exist between agricultural activities in different seasons? Crop Distribution

```
crop_season = pd.crosstab(df['Crop'], df['Season'])  
  
print(crop_season)
```

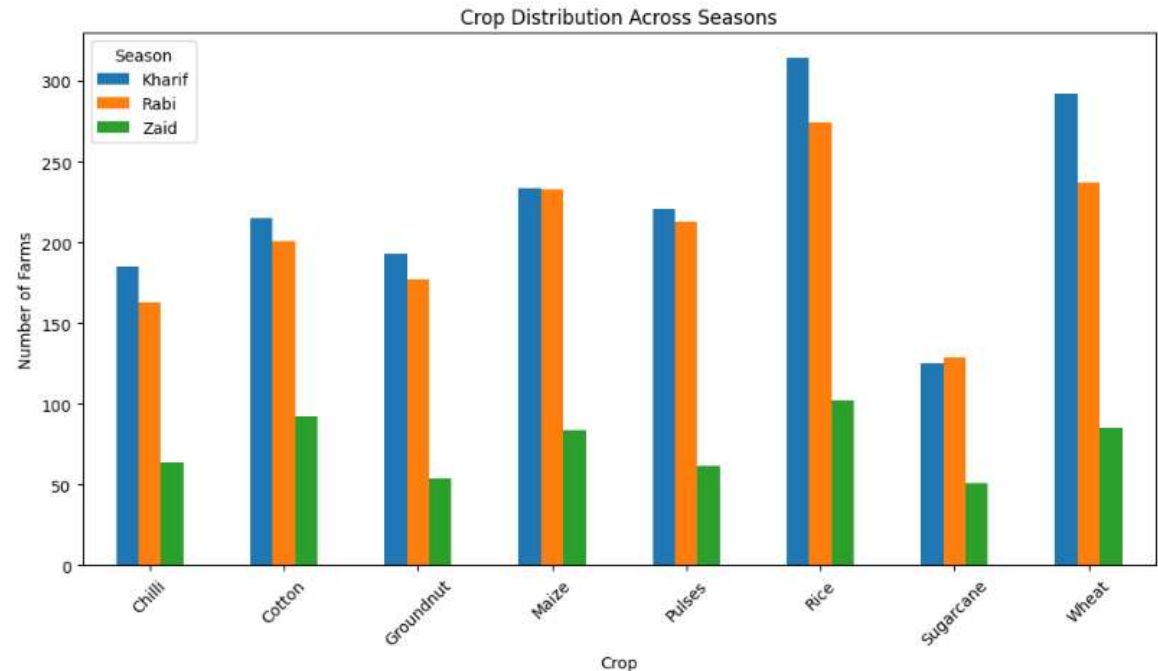
Season	Kharif	Rabi	Zaid
Crop			
Chilli	185	163	64
Cotton	215	201	92
Groundnut	193	177	54
Maize	234	233	84
Pulses	221	213	62
Rice	314	274	102
Sugarcane	125	129	51
Wheat	292	237	85

Visualization:

The visualization shows that agricultural activities are **distributed across all seasons**, with some differences in crop composition and irrigation practices.

Crop distribution and irrigation methods are present in all three seasons. For example, **Rice and Wheat are common crops**, while Flood irrigation is the most frequently used irrigation method across seasons.

```
crop_season.plot(  
    kind='bar',  
    figsize=(12,6),  
    title='Crop Distribution Across Seasons'  
)  
  
plt.xlabel('Crop')  
plt.ylabel('Number of Farms')  
plt.xticks(rotation=45)  
plt.show()
```



RESULTS

5. Are there noticeable variations in resource usage across seasons?

```
resource_usage = df.groupby('Season').agg({
    'Fertilizer_kg_ha': 'mean',
    'Pesticide_Litre_ha': 'mean',
    'Water_Used_m3': 'mean',
    'Water_Efficiency_t_per_1000m3': 'mean'
}).round(2)
```

```
print(resource_usage)
```

Season	Fertilizer_kg_ha	Pesticide_Litre_ha	Water_Used_m3	Water_Efficiency_t_per_1000m3
Kharif	187.08	5.08	6102.20	5.89
Rabi	185.41	5.02	5846.99	5.19
Zaid	184.64	5.17	6419.89	4.41

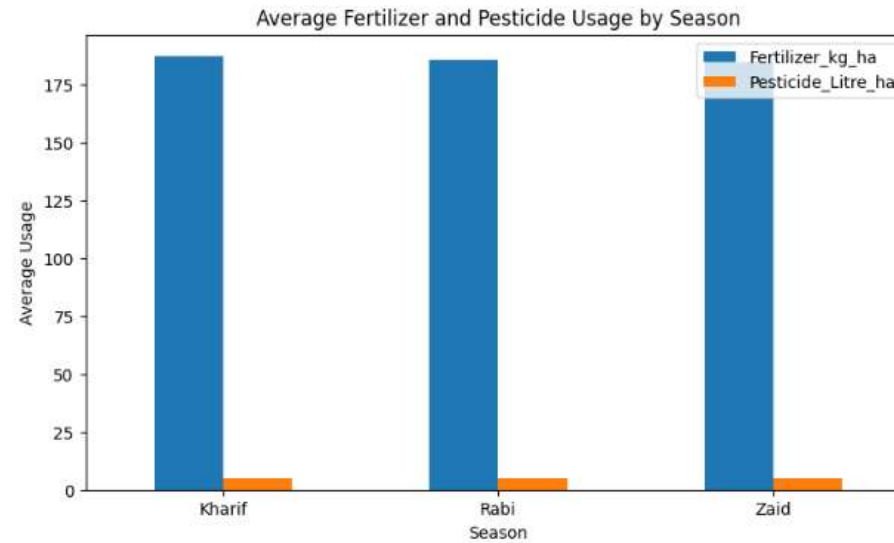
Visualization:

The graph shows that Zaid requires more water despite receiving less rainfall, while fertilizer and pesticide usage remain relatively stable across seasons.

Average water usage is highest in Zaid (6,419.89 m³), followed by Kharif (6,102.20 m³) and Rabi (5,846.99 m³). Fertilizer usage remains relatively similar across seasons.

```
resource_usage[
    ['Fertilizer_kg_ha', 'Pesticide_Litre_ha']
].plot(
    kind='bar',
    figsize=(9,5),
    title='Average Fertilizer and Pesticide Usage by Season'
)

plt.xlabel('Season')
plt.ylabel('Average Usage')
plt.xticks(rotation=0)
plt.show()
```



RESULTS

6. Are there relationships between environmental conditions and agricultural performance?

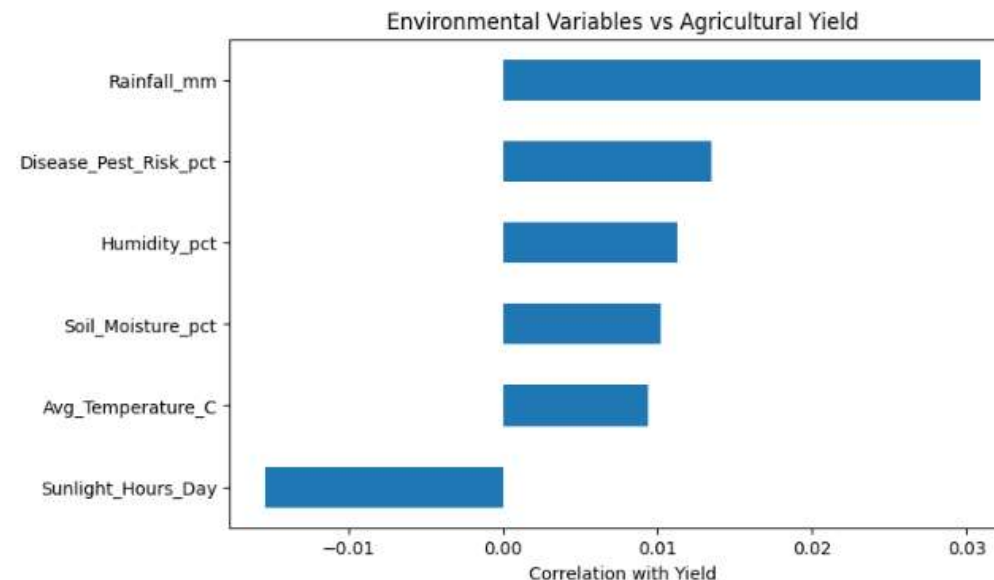
```
environmental_variables = [  
    'Rainfall_mm',  
    'Avg_Temperature_C',  
    'Humidity_pct',  
    'Sunlight_Hours_Day',  
    'Soil_Moisture_pct',  
    'Disease_Pest_Risk_pct'  
]  
  
correlation = df[  
    environmental_variables + ['Yield_Tonnes_Ha']  
].corr()  
  
print(correlation['Yield_Tonnes_Ha'].sort_values(ascending=False))  
  
*** Yield_Tonnes_Ha      1.000000  
    Rainfall_mm      0.030853  
    Disease_Pest_Risk_pct  0.013490  
    Humidity_pct      0.011277  
    Soil_Moisture_pct  0.010210  
    Avg_Temperature_C  0.009349  
    Sunlight_Hours_Day -0.015362  
    Name: Yield_Tonnes_Ha, dtype: float64
```

Visualization:

The visualization shows that seasons with more favorable overall environmental conditions tend to have higher average yields, although the relationship between individual environmental factors and yield is not strong.

The seasonal averages show that Kharif has higher rainfall and soil moisture along with the highest yield. However, the correlation analysis between individual environmental variables and yield is **very weak**, indicating that no strong linear relationship is evident from these variables alone.

```
plt.figure(figsize=(8,5))  
  
correlation['Yield_Tonnes_Ha'].drop('Yield_Tonnes_Ha').sort_values().plot(  
    kind='barh'  
)  
  
plt.title('Environmental Variables vs Agricultural Yield')  
plt.xlabel('Correlation with Yield')  
plt.show()
```



RESULTS

7. How do economic outcomes vary across seasons?

```
economic = df.groupby('Season').agg({
    'Total_Cost_INR': 'mean',
    'Revenue_INR': 'mean',
    'Profit_INR': 'mean'
}).round(2)

print(economic)
```

Season	Total_Cost_INR	Revenue_INR	Profit_INR
Kharif	531804.41	710719.06	178914.65
Rabi	513836.58	601526.05	87689.47
Zaid	543976.73	519171.90	-24804.82

```
profitability = df.groupby('Season').apply(
    lambda x: (x['Profit_INR'] > 0).mean() * 100
)
```

```
print("Percentage of profitable farms:")
print(profitability.round(2))
```

Percentage of profitable farms:

Season	Percentage
Kharif	57.79
Rabi	48.86
Zaid	35.52

dtype: float64

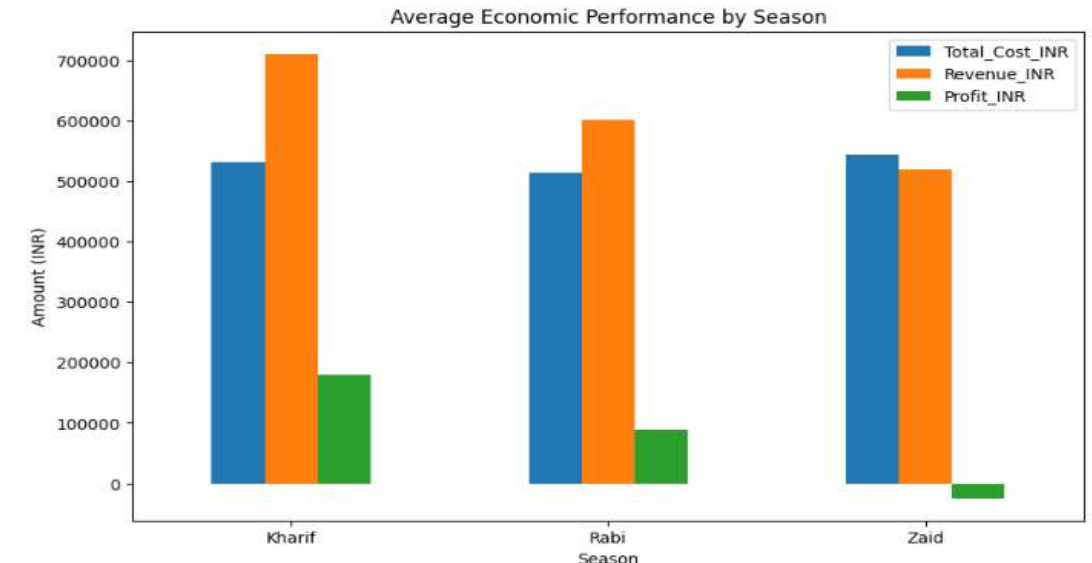
/tmp/ipykernel_473/163162748.py:1: DeprecationWarning: DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated
profitability = df.groupby('Season').apply()

Visualization:

The graph clearly shows **Kharif** as the most economically successful season, while **Zaid** has the weakest economic performance and an average negative profit.

Kharif has the highest average revenue (₹710,719) and profit (₹178,915). Rabi has an average profit of approximately ₹87,689, while Zaid has an average loss of ₹24,805.

```
+economic = df.groupby('Season').agg({
+    'Total_Cost_INR': 'mean',
+    'Revenue_INR': 'mean',
+    'Profit_INR': 'mean'
+}).round(2)
+
+economic.plot(
+    kind='bar',
+    figsize=(10,6),
+    title='Average Economic Performance by Season'
+)
+
+plt.xlabel('Season')
+plt.ylabel('Amount (INR)')
+plt.xticks(rotation=0)
+plt.show()
```



RESULTS

8. Are some seasonal patterns consistent across different regions or categories?

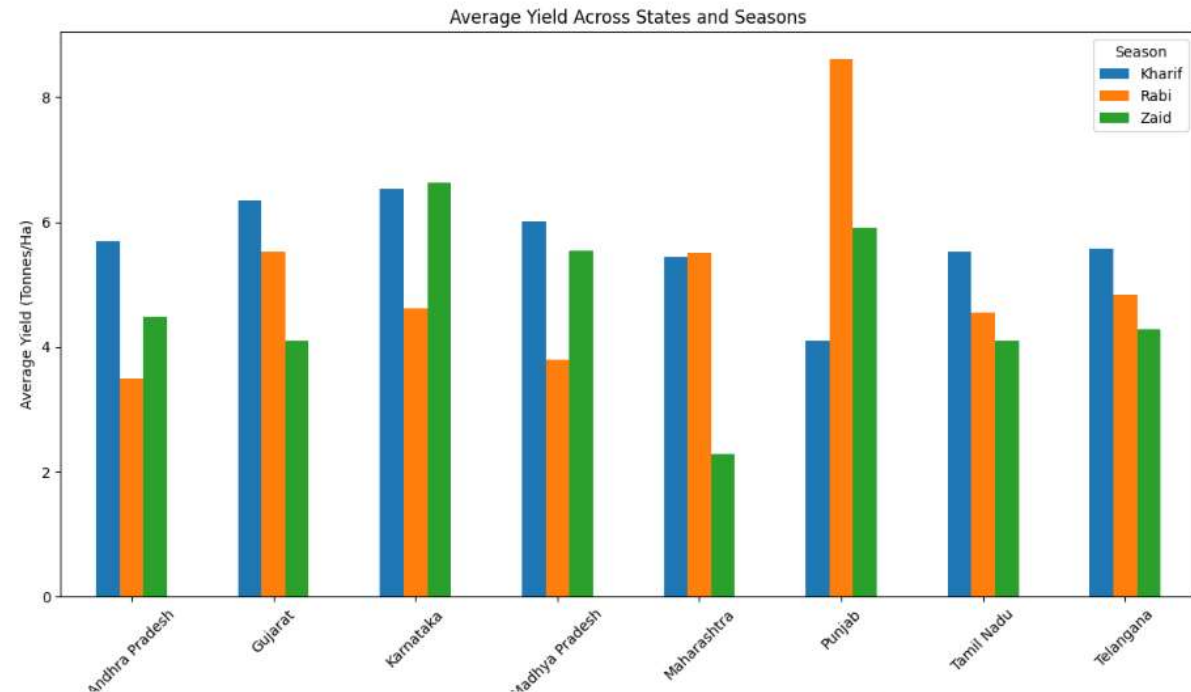
```
state_season = df.groupby(  
    ['State', 'Season']  
)[['Yield_Tonnes_Ha']].mean().unstack()  
  
print(state_season.round(2))
```

Season	Kharif	Rabi	Zaid
State			
Andhra Pradesh	5.69	3.49	4.48
Gujarat	6.34	5.52	4.10
Karnataka	6.52	4.61	6.62
Madhya Pradesh	6.00	3.80	5.54
Maharashtra	5.45	5.50	2.29
Punjab	4.10	8.61	5.91
Tamil Nadu	5.52	4.56	4.10
Telangana	5.57	4.83	4.28

The seasonal performance is **not completely consistent across states**. For example, Punjab has its highest average yield in Rabi (**8.61 tonnes/ha**), whereas Gujarat performs best in Kharif (**6.34 tonnes/ha**), and Karnataka performs slightly better in Zaid (**6.62 tonnes/ha**)

The seasonal performance is **not completely consistent across states**. For example, Punjab has its highest average yield in Rabi (**8.61 tonnes/ha**), whereas Gujarat performs best in Kharif (**6.34 tonnes/ha**), and Karnataka performs slightly better in Zaid (**6.62 tonnes/ha**)

```
state_season.plot(  
    kind='bar',  
    figsize=(14,7),  
    title='Average Yield Across States and Seasons'  
)  
  
plt.xlabel('State')  
plt.ylabel('Average Yield (Tonnes/Ha)')  
plt.xticks(rotation=45)  
plt.show()
```



RESULTS

9. Are there unusual or unexpected seasonal patterns?

```
Q1 = df['Yield_Tonnes_Ha'].quantile(0.25)
Q3 = df['Yield_Tonnes_Ha'].quantile(0.75)

IQR = Q3 - Q1

lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR

outliers = df[
    (df['Yield_Tonnes_Ha'] < lower_limit) |
    (df['Yield_Tonnes_Ha'] > upper_limit)
]

print("Lower Limit:", lower_limit)
print("Upper Limit:", upper_limit)
print("Number of Yield Outliers:", len(outliers))

print("\nOutliers by Season:")
print(outliers['Season'].value_counts())
```

```
... Lower Limit: -1.6349999999999996
Upper Limit: 5.5249999999999995
Number of Yield Outliers: 302
```

```
Outliers by Season:
Season
Kharif    129
Rabi      124
Zaid       49
Name: count, dtype: int64
```

The chart highlights these deviations from the overall seasonal trend and shows that certain regions behave differently from the general pattern.

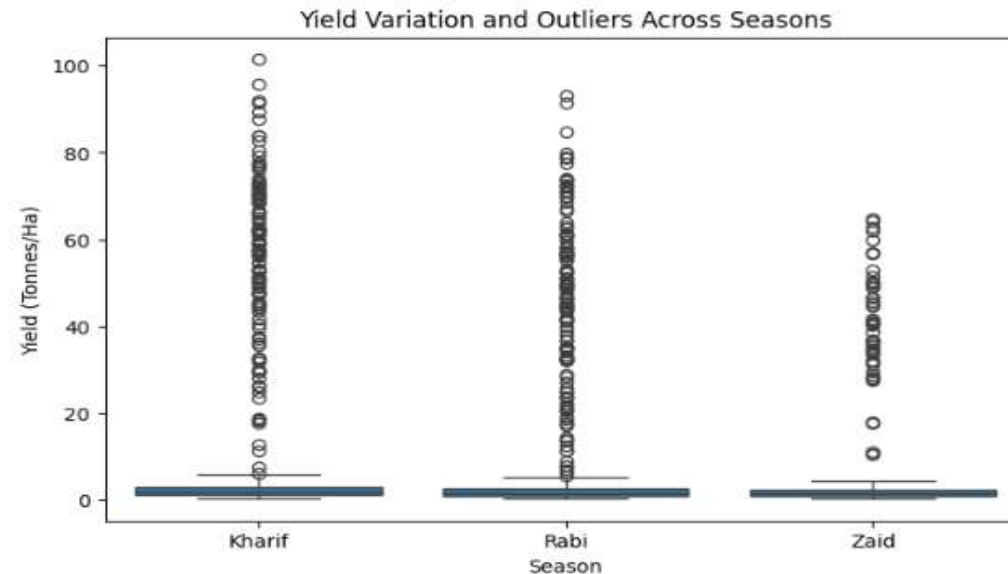
Some unusual regional patterns are visible. For example, Punjab's Rabi yield (8.61 tonnes/ha) is much higher than its Kharif yield (4.10 tonnes/ha), while Karnataka's Zaid yield (6.62 tonnes/ha) slightly exceeds its Kharif yield (6.52 tonnes/ha).

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8, 5))

sns.boxplot(x='Season', y='Yield_Tonnes_Ha', data=df)

plt.title("Yield Variation and Outliers Across Seasons")
plt.xlabel("Season")
plt.ylabel("Yield (Tonnes/Ha)")
plt.show()
```



RESULTS

10. What insights can be derived from the observed seasonal differences?

```
insights = df.groupby('Season').agg([
    'Yield_Tonnes_Ha': 'mean',
    'Revenue_INR': 'mean',
    'Profit_INR': 'mean',
    'Water_Used_m3': 'mean',
    'Water_Efficiency_t_per_1000m3': 'mean',
    'Disease_Pest_Risk_pct': 'mean'
]).round(2)

print(insights)
```

	Yield_Tonnes_Ha	Revenue_INR	Profit_INR	Water_Used_m3	\
Season					
Kharif	5.64	710719.06	178914.65	6102.20	
Rabi	5.08	601526.05	87689.47	5846.99	
Zaid	4.67	519171.90	-24804.82	6419.89	

	Water_Efficiency_t_per_1000m3	Disease_Pest_Risk_pct
Season		
Kharif	5.89	54.47
Rabi	5.19	40.48
Zaid	4.41	38.22

Visualization:

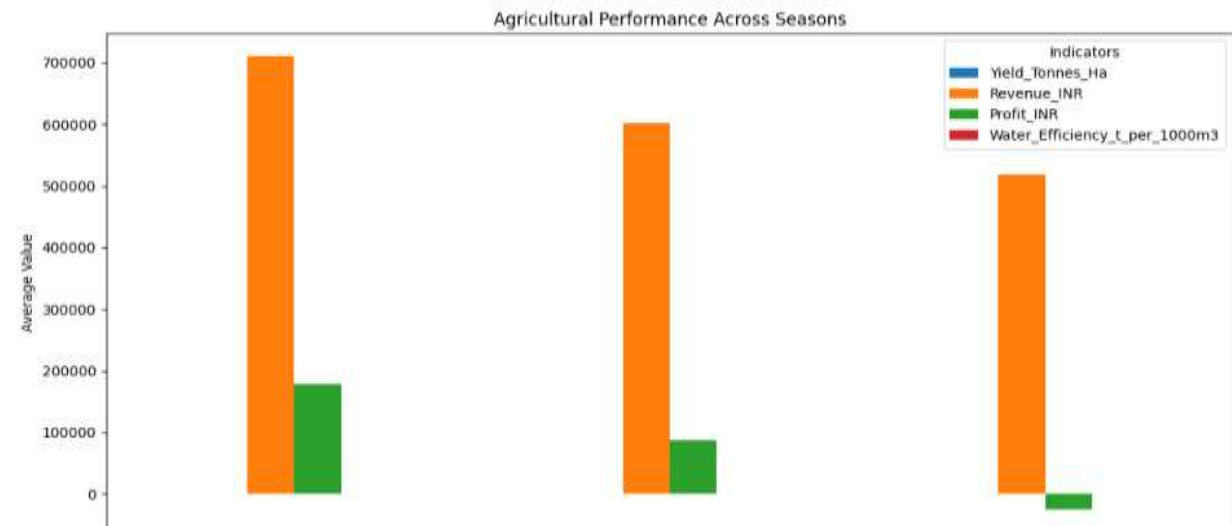
The visualizations make it easy to identify **Kharif** as the **strongest overall season** and **Zaid** as the **weakest**, while also revealing regional and environmental differences.

The analysis indicates that Kharif generally provides the **highest yield, revenue, profit, and water efficiency**, while Zaid has the lowest yield and water efficiency and records negative average profit.

```
plot_data = insights[
    ['Yield_Tonnes_Ha',
     'Revenue_INR',
     'Profit_INR',
     'Water_Efficiency_t_per_1000m3']
]

plot_data.plot(
    kind='bar',
    figsize=(12, 6)
)

plt.title("Agricultural Performance Across Seasons")
plt.xlabel("Season")
plt.ylabel("Average Value")
plt.xticks(rotation=0)
plt.legend(title="Indicators")
plt.tight_layout()
plt.show()
```



RESULTS

11. What conclusions can reasonably be drawn from the available data?

```
from scipy.stats import f_oneway

kharif = df[df['Season'] == 'Kharif']['Yield_Tonnes_Ha'].dropna()
rabi = df[df['Season'] == 'Rabi']['Yield_Tonnes_Ha'].dropna()
zaid = df[df['Season'] == 'Zaid']['Yield_Tonnes_Ha'].dropna()

statistic, p_value = f_oneway(kharif, rabi, zaid)

print("ANOVA F-statistic:", statistic)
print("ANOVA p-value:", p_value)

if p_value < 0.05:
    print("The seasonal yield differences are statistically significant.")
else:
    print("The seasonal yield differences are not statistically significant.")
```

```
*** ANOVA F-statistic: 1.4579589466621523
ANOVA p-value: 0.23283549440109827
The seasonal yield differences are not statistically significant.
```

Visualization:

The charts provide a clear overall picture of seasonal differences in agricultural, environmental, resource, and economic performance, supporting the numerical findings.

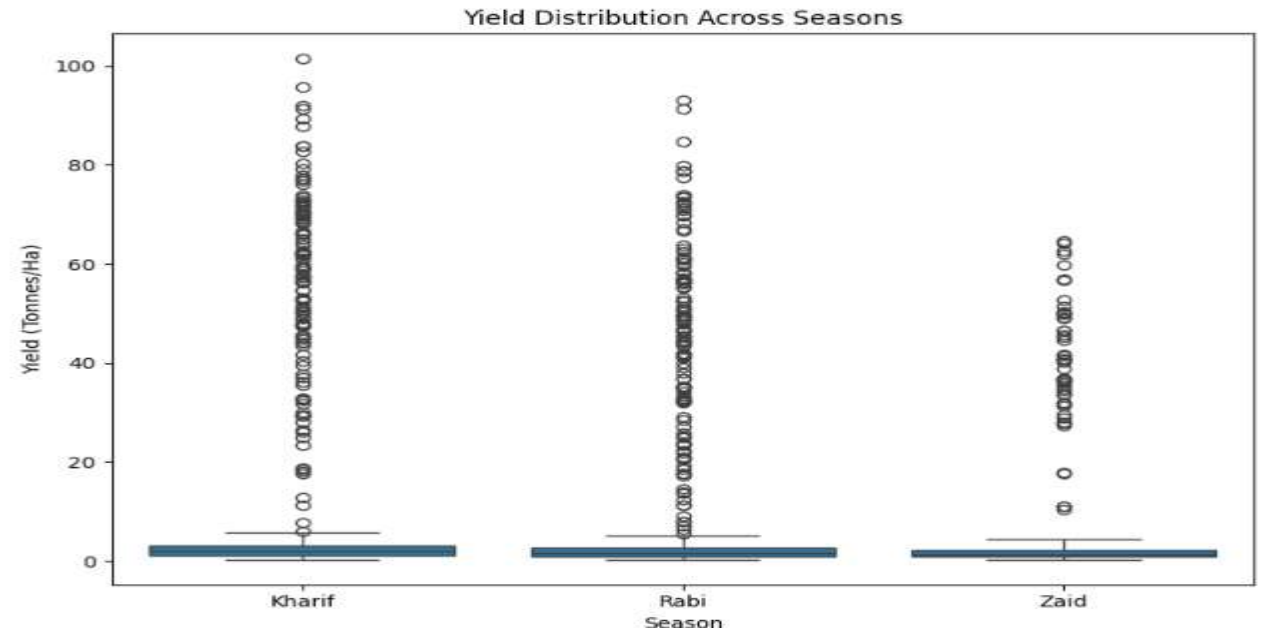
The analysis suggests that agricultural performance varies substantially by season. Kharif has the strongest overall performance, while Zaid faces lower yield, higher water usage, lower water efficiency, and negative average profit.

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8, 6))

sns.boxplot(
    x='Season',
    y='Yield_Tonnes_Ha',
    data=df
)

plt.title("Yield Distribution Across Seasons")
plt.xlabel("Season")
plt.ylabel("Yield (Tonnes/Ha)")
plt.tight_layout()
plt.show()
```



RESULTS

12. How could the findings support better seasonal agricultural planning?

```
planning = df.groupby('Season').agg({
    'Yield_Tonnes_Ha': 'mean',
    'Profit_INR': 'mean',
    'Water_Used_m3': 'mean',
    'Water_Efficiency_t_per_1000m3': 'mean',
    'Disease_Pest_Risk_pct': 'mean'
}).round(2)

print(planning)
```

Season	Yield_Tonnes_Ha	Profit_INR	Water_Used_m3	Water_Efficiency_t_per_1000m3	Disease_Pest_Risk_pct
Kharif	5.64	178914.65	6102.20	5.89	54.47
Rabi	5.08	87689.47	5846.99	5.19	40.48
Zaid	4.67	-24804.82	6419.89	4.41	38.22

The visualization compares the key indicators across **Kharif, Rabi, and Zaid** seasons. It makes seasonal differences easier to understand and shows how the findings can be used for more effective agricultural planning.

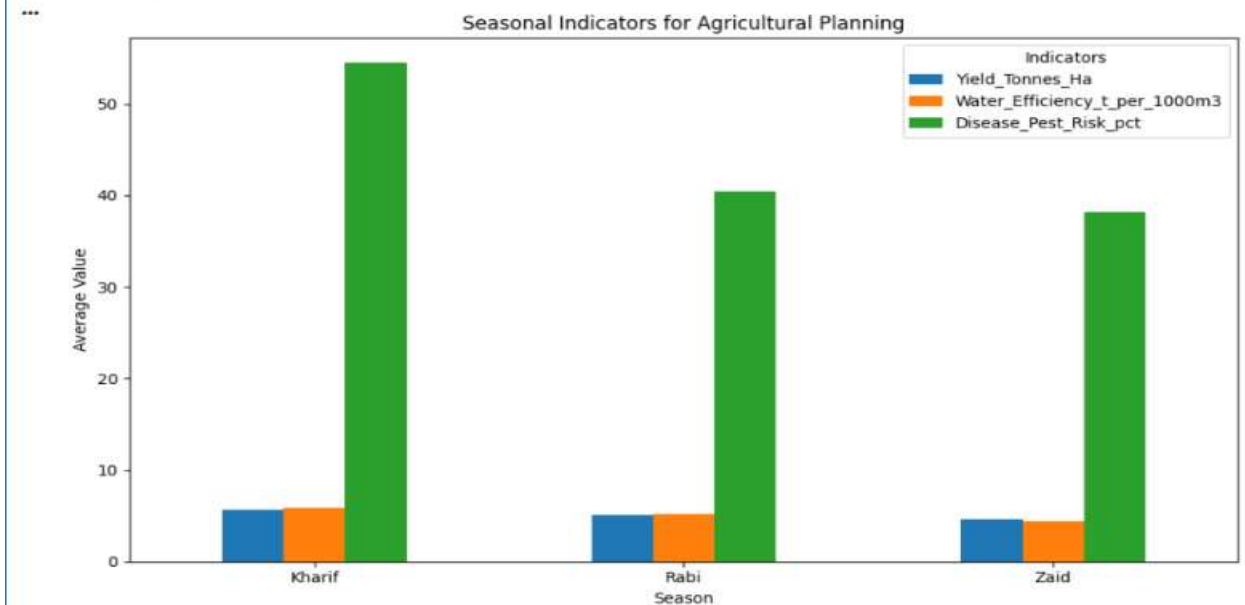
Code / Output:

The results can help farmers and planners identify seasons with better productivity and profitability, while paying particular attention to **water management and economic risks during Zaid**.

```
planning_data = insights[
    ['Yield_Tonnes_Ha',
     'Water_Efficiency_t_per_1000m3',
     'Disease_Pest_Risk_pct']
]

planning_data.plot(
    kind='bar',
    figsize=(10, 6)
)

plt.title("Seasonal Indicators for Agricultural Planning")
plt.xlabel("Season")
plt.ylabel("Average Value")
plt.xticks(rotation=0)
plt.legend(title="Indicators")
plt.tight_layout()
plt.show()
```



Future scope

- 1. Predictive Analysis** - Future work can use machine learning to predict crop yield, profit, and agricultural performance for different seasons.
- 2. Real-Time Data Integration** - Weather forecasts, rainfall, soil moisture, and other real-time environmental data can be incorporated to improve seasonal decisions.
- 3. Crop Recommendation** - The analysis can be extended to recommend suitable crops for each season based on yield, water requirements, soil conditions, and expected profit.
- 4. Regional-Level Planning** - Future analysis can study seasonal patterns across different geographical regions and provide region-specific agricultural recommendations.
- 5. Resource Optimization** - Water, fertilizer, pesticide, and other resource usage can be optimized to improve productivity while reducing unnecessary consumption.
- 6. Early Risk Detection** - Disease and pest risk data can be used to develop early-warning systems for farmers.
- 7. Interactive Dashboard** - The project can be developed into an interactive dashboard where users can select a season or region and view agricultural performance and recommendations

GitHub Link

[GitHub link](#)

<https://github.com/25A31A42H5/Seasonal-Agriculture-Performance-Analysis>.

VOIS Course completion certificate -

Course name- Data Visualization



Thank you

I sincerely thank you for taking the time to review my project.
I am grateful for your valuable feedback, guidance, and suggestions, which will help me improve my work and learning.

Thank you for your time and support.